

# Explainable Deep Reinforcement Learning for Multi-Asset Trading via Transformer-Based State Encoding

## Graduation Thesis

Student Name: LE NHAT QUYEN

Student ID: MAMAIU21100

Thesis Supervisor: ASSOC. PROF. NGUYEN MINH QUAN

Department of Mathematics  
International University, VNU-HCM

June 2026

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# 1. Introduction

## Background and Motivation

### 01 **Portfolio-Level Decision Making**

The main question is how to allocate capital across several assets, not how to predict each stock separately.

### 02 **Changing Market Conditions**

Market behavior changes over time, so a fixed allocation may become less suitable.

### 03 **Dynamic Allocation under Transaction cost**

Portfolio weights should adapt to new information, but frequent rebalancing creates transaction costs.

**Motivation:** Learning an adaptive portfolio allocation policy from historical multi-asset data under transaction costs.

# 1. Introduction

## From Prediction to Portfolio Allocation

### Prediction View

- ▶ Predict price, return, or direction
- ▶ Asset-level output
- ▶ Supervised forecasting loss
- ▶ Accuracy, RMSE, directional accuracy

vs

### Allocation View

- ▶ Generate portfolio weights
- ▶ Portfolio-level action
- ▶ Reinforcement learning reward
- ▶ Return, Sharpe, drawdown, turnover, cost

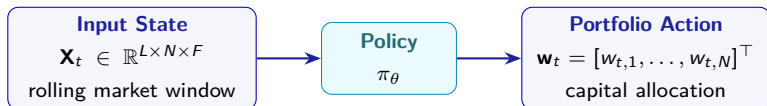
**Key distinction:** This thesis shifts from separate stock-price forecasting to direct learning of portfolio allocation behavior from portfolio outcomes.

# 1. Introduction

## Problem Setting

### Dynamic Portfolio Allocation

At each decision time  $t$ , the model observes a recent market window and produces a portfolio allocation for the next trading period.



► **Sequential nature:** Current allocation affects future portfolio value and future rebalancing cost.

# 1. Introduction

## Research Objectives and Questions

### Research Objectives

- ▶ Investigate the role of state representation learning in DRL portfolio allocation.
- ▶ Compare MLP, LSTM, and Transformer encoders under a common PPO framework.
- ▶ Examine how transaction-cost-aware rewards affect performance and behavior.
- ▶ Use temporal attention as supporting interpretability evidence.

### Research Questions

**RQ1.** How does the choice of state encoder-MLP, LSTM, or Transformer affect PPO-based portfolio allocation performance under a common PPO framework?

**RQ2.** How does incorporating a transaction-cost penalty into the reward function influence portfolio performance and policy behavior?

**Supporting.** What do temporal attention patterns suggest about historical information usage?

**Presentation flow:** Allocation Problem → Proposed Framework → Empirical Answers.

## 2. Literature Review

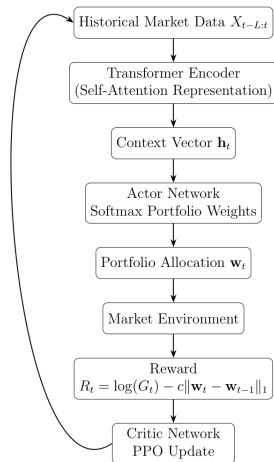
### Main Research Streams

| Research Stream            | Key References                                  | Relevance to This Thesis                                                                    |
|----------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------|
| Portfolio optimization     | Markowitz (1952); Sharpe (1964)                 | Provides portfolio-level return–risk foundations and motivates allocation-based evaluation. |
| Financial machine learning | Gu et al. (2020); Heaton et al. (2017)          | Supports learning nonlinear patterns from noisy financial market data.                      |
| Attention-based models     | Vaswani et al. (2017); Gezici and Sefer (2024)  | Supports market-state encoding through self-attention and sequence modeling.                |
| Reinforcement learning     | Sutton and Barto (2018); Schulman et al. (2017) | Provides the state–action–reward framework for sequential allocation decisions.             |
| Cost-aware trading         | Moody and Saffell (2001); Jiang et al. (2017)   | Motivates cost-aware reward design and interpretation of rebalancing behavior.              |

**Research gap:** Prior work often treats forecasting, allocation, costs, and interpretability separately; this thesis examines them together in a controlled portfolio-allocation framework.

# 3. Proposed Methodology

## Proposed Transformer-PPO Framework



### Pipeline Logic

- ▶ Historical market data are encoded by the Transformer.
- ▶ The latent state is passed to a PPO actor-critic policy.
- ▶ The actor outputs portfolio weights.
- ▶ Reward reflects portfolio growth and transaction cost.

$$X_{t-L:t} \rightarrow h_t \rightarrow w_t \rightarrow G_t \rightarrow R_t \rightarrow \text{PPO update}$$

Figure 3.1. Proposed Transformer-PPO framework for dynamic portfolio allocation.



# 3. Proposed Methodology

## MDP Formulation: State, Action, and Reward

### MDP Components

|                 |                                                                                                       |                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| ► <b>State</b>  | $\mathbf{x}_t \in \mathbb{R}^{L \times N \times F}$                                                   | Rolling market tensor                             |
| ► <b>Action</b> | $\mathbf{w}_t = [w_{t,1}, \dots, w_{t,N}]^\top$<br>$w_{t,i} \geq 0, \quad \sum_{i=1}^N w_{t,i} = 1$   | Portfolio weights<br>Long-only, fully invested.   |
| ► <b>Reward</b> | $G_t = \mathbf{w}_t^\top \mathbf{y}_t$<br>$R_t = \log(G_t) - c \ \mathbf{w}_t - \mathbf{w}_{t-1}\ _1$ | Portfolio growth<br>Growth minus rebalancing cost |

**Interpretation:** The agent maps recent market information into feasible portfolio weights and learns from realized portfolio outcomes with a rebalancing-cost penalty.

# 3. Proposed Methodology

## Transformer State Encoder

### Transformer State Encoder

|                       |                                                        |                                                  |
|-----------------------|--------------------------------------------------------|--------------------------------------------------|
| ► <b>Latent State</b> | $\mathbf{h}_t = f_{\psi}(\mathbf{X}_t)$                | Compact market representation                    |
| ► <b>Attention</b>    | $\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$ | Nonuniform weighting across historical positions |
| ► <b>Role</b>         | Transformer encoder                                    | Supplies temporal market state to PPO            |
| ► <b>Comparison</b>   | MLP / LSTM / Transformer                               | Same PPO framework for encoder comparison        |

**Interpretation:** The encoder maps the rolling market window into a latent state representation, allowing the PPO policy to learn from temporally weighted market information.

# 3. Proposed Methodology

## Actor–Critic Portfolio Policy

### Actor–Critic Components

|                            |                                               |                                                        |
|----------------------------|-----------------------------------------------|--------------------------------------------------------|
| ► <b>Actor</b>             | $\mathbf{z}_t = g_{\theta}(\mathbf{h}_t)$     | Maps encoded market state into allocation scores       |
| ► <b>Portfolio Weights</b> | $\mathbf{w}_t = \text{softmax}(\mathbf{z}_t)$ | Ensures long-only and fully invested allocation        |
| ► <b>Critic</b>            | $V_t = V_{\phi}(\mathbf{h}_t)$                | Estimates state value for PPO learning                 |
| ► <b>PPO Update</b>        | Actor–critic PPO                              | Actor decides allocation; critic supports optimization |

**Interpretation:** The actor converts the encoded market state into feasible portfolio weights, while the critic estimates state value to support PPO policy optimization.

# 3. Proposed Methodology

## Cost-Aware Reward Function

### Reward Components

|                            |                                                                            |                                                             |
|----------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|
| ► <b>Portfolio Growth</b>  | $G_t = \mathbf{w}_t^\top \mathbf{y}_t$                                     | Portfolio growth from selected weights and realized returns |
| ► <b>Transaction Cost</b>  | $TC_t = c \ \mathbf{w}_t - \mathbf{w}_{t-1}\ _1$                           | Turnover-based cost from rebalancing                        |
| ► <b>Cost-Aware Reward</b> | $R_t = \log(G_t) - TC_t$                                                   | Balances portfolio growth and trading cost                  |
| ► <b>Reward Objective</b>  | $R_t = \log(\mathbf{w}_t^\top \mathbf{y}_t) - c \ \Delta \mathbf{w}_t\ _1$ | Growth minus rebalancing penalty                            |

**Interpretation:** The reward encourages portfolio growth while penalizing excessive rebalancing, so the policy learns under transaction-cost-aware trading behavior.

# 3. Proposed Methodology

## PPO Optimization and Controlled Encoder Comparison

### PPO clipped objective

$$L^{\text{clip}}(\theta) = \mathbb{E}_t \left[ \min \left( \rho_t(\theta) \hat{A}_t, \text{clip} \left( \rho_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right]$$

### Role in This Thesis

- ▶ PPO updates the actor and critic from collected portfolio trajectories.
- ▶ Clipping prevents overly aggressive policy updates.
- ▶ PPO is fixed across MLP, LSTM, and Transformer variants.

**Takeaway:** Keeping PPO fixed helps isolate the role of state-encoder choice in learned-policy performance.

## 4. Experimental Design and Implementation

### Data Source and Dataset Description

#### ► U.S. Multi-Asset Equity Portfolio (2020–2024)

| Ticker | Company Name         | Profile                       |
|--------|----------------------|-------------------------------|
| AAPL   | Apple Inc.           | Large-cap technology          |
| MSFT   | Microsoft Corp.      | Large-cap technology          |
| AMZN   | Amazon.com Inc.      | Tech / consumer discretionary |
| JPM    | JPMorgan Chase & Co. | Financial services            |
| JNJ    | Johnson & Johnson    | Healthcare                    |

### Data Characteristics:

- **Period:** 2020–2024
- **Source:** Yahoo Finance via `yfinance`
- **Benchmark:** Equal Weight; Buy-and-Hold Equal Weight
- **Frequency:** Daily

# 4. Experimental Design and Implementation

## Controlled Experimental Setup

### ► Controlled Portfolio Allocation Experiment

| Component        | Setting                          |
|------------------|----------------------------------|
| Asset universe   | Five U.S. equities               |
| State window     | 20 trading days                  |
| Portfolio rule   | Long-only, fully invested        |
| Action output    | Portfolio weights $\mathbf{w}_t$ |
| Optimizer        | PPO actor-critic                 |
| Encoder variants | MLP, LSTM, Transformer           |
| Reward variants  | Cost-aware vs. no-cost           |
| Model selection  | Best validation checkpoint       |
| Test protocol    | Out-of-sample test               |

*Validation selects checkpoints; the chronological test set is used only for final out-of-sample evaluation.*

### Chronological Split

|            |           |
|------------|-----------|
| Training   | 717 steps |
| Validation | 250 steps |
| Test       | 251 steps |

### Controlled Factors

- Same asset universe
- Same portfolio constraints
- Same PPO learning rule
- Same evaluation protocol

### Compared Factors

- State encoder design
- Reward specification

# 4. Experimental Design and Implementation

## Portfolio Evaluation Metrics

### ► Evaluation of Portfolio Performance

| Dimension                 | Metrics                                  |
|---------------------------|------------------------------------------|
| Return                    | Final portfolio value; cumulative return |
| Risk                      | Daily volatility; maximum drawdown       |
| Risk-adjusted performance | Sharpe approximation                     |
| Trading behavior          | Average turnover                         |
| Cost exposure             | Total and average trading cost           |

### Main Evaluation Logic

- Return measures portfolio growth.
- Risk captures volatility and downside loss.
- Turnover reflects rebalancing behavior.

#### Evaluation Principle

Portfolio performance is interpreted through **return**, **risk**, **turnover**, and **trading cost** jointly.



# 5. Empirical Results and Discussion

## Overall Out-of-Sample Results

| Model           | Cum. Return | Sharpe | MDD     | Avg. Turnover | Rank |
|-----------------|-------------|--------|---------|---------------|------|
| Equal Weight    | 0.2501      | 1.7390 | -0.0888 | 0.0000        | 1    |
| Buy-and-Hold EW | 0.2492      | 1.6802 | -0.0967 | 0.0000        | 2    |
| Transformer-PPO | 0.1947      | 1.5747 | -0.1052 | 0.2165        | 3    |
| LSTM-PPO        | 0.1692      | 1.5377 | -0.0911 | 0.1913        | 4    |
| MLP-PPO         | 0.1212      | 1.4207 | -0.0968 | 0.2679        | 5    |
| Random Policy   | 0.0008      | 1.1497 | -0.1049 | 0.6311        | 6    |

### 01 Overall Result

Passive benchmarks show the strongest overall out-of-sample performance.

### 02 Learned Policy Result

Transformer-PPO gives the best learned-policy performance.

### 03 Careful Interpretation

Transformer-PPO is strongest within learned PPO models, but does not dominate passive benchmarks.

**Main takeaway:** Passive benchmarks remain strongest overall, while Transformer-PPO gives the best learned-policy result.

# 5. Empirical Results and Discussion

## Portfolio Value Curves

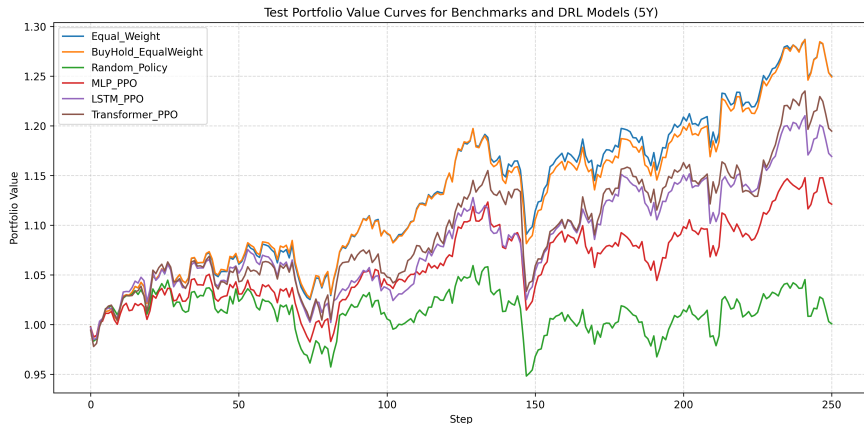


Figure 5.1. Out-of-sample portfolio value curves across benchmarks and learned PPO models.

**Interpretation:** Passive benchmarks finish highest; Transformer-PPO leads the learned PPO group.

# 5. Empirical Results and Discussion

## Main RQ: Encoder Comparison

| Model                  | Cum. Return   | Sharpe        | MDD     | Avg. Turnover | Rank     |
|------------------------|---------------|---------------|---------|---------------|----------|
| <b>Transformer-PPO</b> | <b>0.1947</b> | <b>1.5747</b> | -0.1052 | 0.2165        | <b>1</b> |
| LSTM-PPO               | 0.1692        | 1.5377        | -0.0911 | 0.1913        | 2        |
| MLP-PPO                | 0.1212        | 1.4207        | -0.0968 | 0.2679        | 3        |

### 01 Evidence

Transformer-PPO leads the learned PPO group.

### 02 Interpretation

State-representation choice is associated with differences in learned-policy performance.

### 03 Boundary

This finding is limited to the learned-model comparison.

**Empirical answer to RQ1:** Under the same PPO framework, encoder choice matters; Transformer-PPO performs best among the tested learned PPO variants.

# 5. Empirical Results and Discussion

## Main RQ: Encoder Comparison

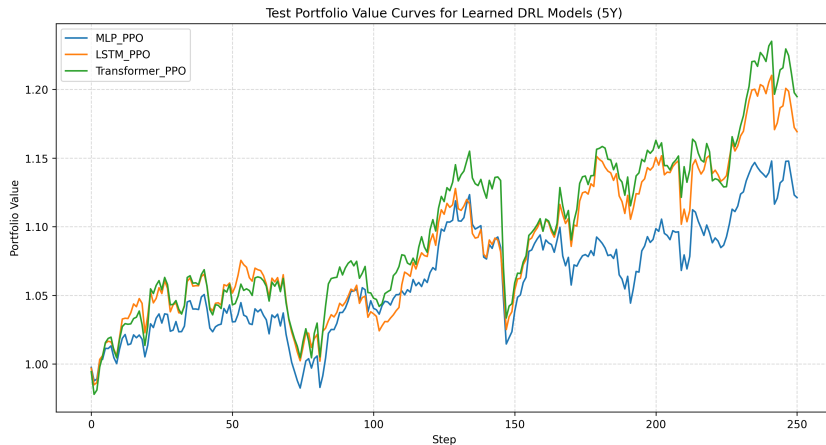


Figure 5.2. Learned PPO encoder comparison based on cumulative return and Sharpe approximation.

**Takeaway:** Transformer-PPO gives the best learned-policy result among the encoder variants.

# 5. Empirical Results and Discussion

## RQ2: Reward-Design Comparison

| Metric             | Cost-Aware | No-Cost |
|--------------------|------------|---------|
| Cumulative Return  | 0.1947     | 0.1898  |
| Sharpe Approx.     | 1.5747     | 1.8691  |
| Maximum Drawdown   | -0.1052    | -0.0614 |
| Average Turnover   | 0.2165     | 0.1790  |
| Total Trading Cost | 0.0544     | 0.0449  |

### 01 Evidence

Cost-aware reward gives slightly higher cumulative return.

### 02 Interpretation

Reward design changes both portfolio performance and trading behavior.

### 03 Boundary

The cost-aware reward is not uniformly better across all evaluation metrics.

**Empirical answer to RQ2:** Transaction-cost-aware reward affects portfolio performance and policy behavior, but its effect is metric-dependent.

# 5. Empirical Results and Discussion

## Portfolio Allocation Behavior

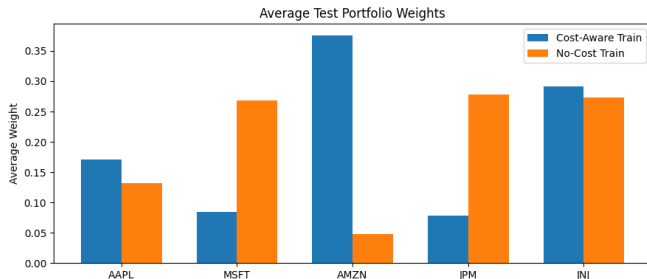


Figure 5.3. Average test-period portfolio allocation under cost-aware and no-cost rewards.

### 01 Evidence

Reward design changes average portfolio composition.

### 02 Cost-aware

Cost-aware policy concentrates more on **AMZN** and **JNJ**.

### 03 No-cost

No-cost policy allocates more to **MSFT** and **JPM**.

**Interpretation:** Reward design changes not only performance metrics, but also learned allocation behavior.

# 5. Empirical Results and Discussion

## Attention Analysis for Interpretability

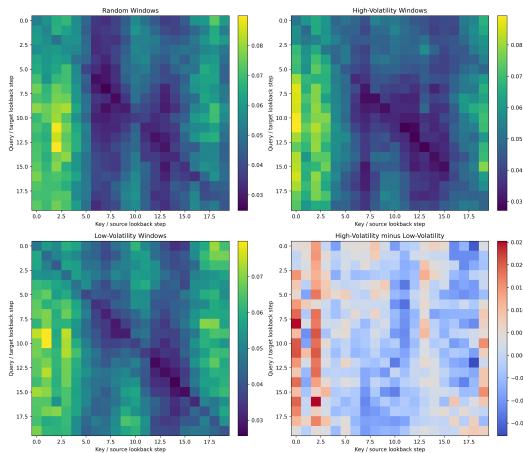


Figure 5.4. Grouped temporal attention patterns across different market conditions.

### Observed Patterns

- ▶ Attention is structured and nonuniform across the lookback window.
- ▶ High-volatility windows show broader temporal emphasis.
- ▶ Low-volatility windows are slightly more recent-focused.

### Important Limitation

Attention is **diagnostic evidence at the representation level**, not a causal explanation of individual portfolio weights.

**Interpretability result:** Attention reveals structured and nonuniform temporal information usage across market conditions, but not causal attribution of portfolio weights.

# 5. Empirical Results and Discussion

## Integrated Findings

### ► Synthesis of Empirical Findings

| Dimension        | Empirical Finding                            | Research Implication                                         |
|------------------|----------------------------------------------|--------------------------------------------------------------|
| Representation   | Transformer-PPO is the best learned policy.  | Transformer encoding supports learned-policy performance.    |
| Benchmark        | Passive benchmarks remain strongest overall. | Passive baselines are essential.                             |
| Reward Design    | Reward effects are metric-dependent.         | Reward design changes behavior and metrics.                  |
| Interpretability | Attention patterns are structured, not flat. | Attention supports interpretability, not causal explanation. |

**Core contribution:** Controlled evidence on how state representation, reward design, and attention analysis support DRL portfolio allocation under transaction costs.



# 6. Conclusion

## Limitations and Future Work

### From empirical scope to future research directions

| Current Scope / Limitation         |   | Future Extension                  |
|------------------------------------|---|-----------------------------------|
| Small five-stock universe          | ⇒ | Larger asset universes            |
| Simplified transaction-cost model  | ⇒ | Realistic trading frictions       |
| No max-weight or sector constraint | ⇒ | Portfolio and risk constraints    |
| PPO as the only RL optimizer       | ⇒ | Alternative DRL algorithms        |
| Single-seed training               | ⇒ | Multi-seed robustness checks      |
| Attention as diagnostic evidence   | ⇒ | Stronger interpretability methods |

**Conclusion:** These limitations define the empirical scope and motivate future extensions.

# 6. Conclusion

## Final Remarks

### ► Representation Learning Result

**Transformer-PPO** performs best among learned PPO variants under the controlled framework.

### ► Reward Design

Cost-aware reward changes allocation behavior, but its effect is **metric-dependent**.

### ► Scope & Interpretability

Passive benchmarks remain strong; attention analysis provides **diagnostic evidence**, not causal attribution.

**Final takeaway:** Among MLP-, LSTM-, and Transformer-based PPO variants, **Transformer-PPO** performs best under the controlled framework.

## 6. Conclusion

Question and Answer

# THANK YOU!

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*I look forward to your questions and feedback.*

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# Appendix

## Ten-Year Robustness Check

| Model                          | Cum. Return | Sharpe | MDD     | Avg. Turnover | Rank |
|--------------------------------|-------------|--------|---------|---------------|------|
| Buy-and-Hold EW                | 0.7807      | 1.9359 | -0.1136 | 0.0000        | 1    |
| Equal Weight                   | 0.7242      | 1.9912 | -0.1020 | 0.0000        | 2    |
| Transformer-PPO 10Y No-Cost    | 0.6451      | 1.9041 | -0.1101 | 0.1467        | 3    |
| MLP-PPO 10Y                    | 0.6363      | 2.2136 | -0.1040 | 0.2279        | 4    |
| LSTM-PPO 10Y                   | 0.6028      | 1.8763 | -0.1077 | 0.1467        | 5    |
| Transformer-PPO 10Y Cost-Aware | 0.5627      | 2.0576 | -0.1237 | 0.2719        | 6    |
| Random Policy                  | 0.2960      | 2.0132 | -0.1580 | 0.6416        | 7    |

### 01 Evidence

Passive benchmarks remain strongest in the extended horizon.

### 02 Learned Policy Result

Transformer-PPO No-Cost is the strongest learned model.

### 03 Boundary

This is supplementary robustness evidence, not the main empirical conclusion.

**Appendix takeaway:** The ten-year check supports a cautious interpretation: passive benchmarks remain difficult to beat, while learned-policy rankings are sensitive to horizon and reward design.

# Appendix

## Ten-Year Portfolio Value Curves

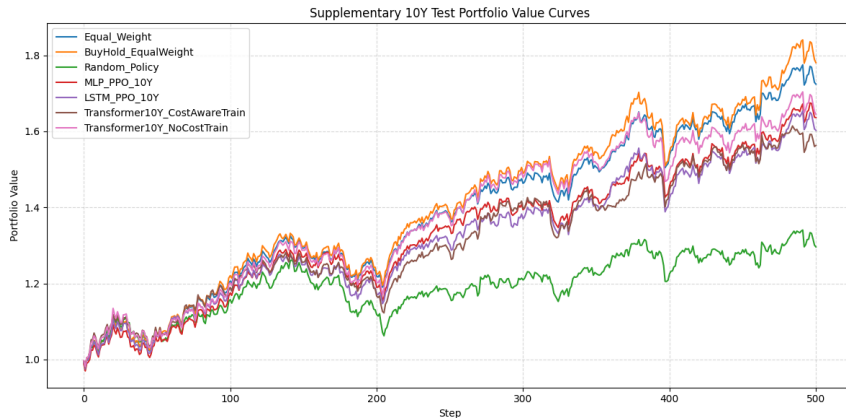


Figure A1. Supplementary ten-year portfolio value curves.

**Interpretation:** The curve is used only as supplementary robustness evidence to inspect whether performance gaps persist over a longer evaluation horizon.